The Hitchhiker's Guide to Mechanistic Interpretability of Vision-Language-Action Models

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Abstract

The mechanistic interpretability toolkit for large language models (LLMs) was built around properties that autoregressive text transformers happen to have: one residual stream, one unembedding matrix, homogeneous layers, a single forward pass per output, a verifiable target token. These are not laws of computation but conveniences of one architectural choice. As the field moves from LLMs to Vision-Language-Action (VLA) models, between two and six of these conveniences quietly disappear depending on the VLA's design, and the tools that depend on them keep producing numerically plausible outputs that no longer mean what they did on an LLM. We name seven such conveniences as cautions **A1–A7**, classify VLAs into seven architecture types, and identify which caution each type breaks. The result is a compact map of where the existing LLM-style toolkit is safe to apply, where it requires a small adaptation, and where it cannot be made to work without re-engineering.

1. Introduction

Mechanistic interpretability seeks to reverse-engineer neural networks at the level of individual components: which attention heads implement which computations, how information flows through the residual stream, and which minimal circuit is causally responsible for a given behaviour (Elhage et al., 2021; Olsson et al., 2022). Progress on language models has been substantial, with circuits identified for indirect object identification, factual recall, and induction (Wang et al., 2023; Meng et al., 2022), and feature-level analyses scaling via sparse autoencoders (Cunningham et al., 2023; Bricken et al., 2023).

Vision-Language-Action (VLA) models now lead generalist robot control: RT-2 (Brohan et al., 2023), OpenVLA (Kim et al., 2024), π_0 (Black et al., 2024), and Octo (Octo Project Team, 2023) couple a language-model backbone with a robot action head and produce motor commands in a single differentiable system. The safety stakes are high: a robot acting on a misunderstood scene causes physical harm in ways a language model Producing a wrong token does not.

VLA mechanistic interpretability is, however, almost entirely unexplored, and we argue that the tooling and techniques don't translate directly and infact it's a harder problem. The architectural diversity of VLAs means the LLM interpretability toolkit, applied without modification, ranges from fully applicable to silently and systematically incorrect depending on which VLA is studied. Applying Direct Logit Attribution (DLA) to a cross-attention fusion VLA gets a heatmap that looks plausible and is wrong — the cross-attention contributions are missing entirely, attributed to nothing. Applying SVD to a continuous-action VLA's residual stream identifies heads with zero causal control over the output while missing the heads that drive it. Neither tool raises an error.

This is a positioning paper aimed at the practitioner about to run a mech interp experiment on a VLAs. We check how much of the mech interp tooling and techniques translate to VLAs. We name the LLM-side properties the toolkit silently depends on, identify which VLA architectures preserve or remove each property, and describe the failure modes that follow.

2. Related Work

Häon et al. (2025) are the closest prior work in spirit. They apply LLM-style mech interp to OpenVLA and π_0 by projecting FFN activations onto the token embedding space, recover sparse "semantic" directions like "slow", and use these as inference-time steering vectors, reporting that under 25% of FFN neurons are "rewired" for action prediction.

Grant et al. (2026) did the first cross-VLA mech-interp study: six models, about 394k rollouts, using SAEs, linear probes, and activation injection. Their main findings are that vision contributes more than language, that language matters more on some tasks than others, and that the network splits into separate motor and goal pathways. They also note that

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activations have to be read one token at a time — averaging across the sequence destroys the signal. That last point is exactly what our **A3/A6** cautions predict, and we treat it as outside confirmation.

3. Background: The LLM Toolkit

A transformer with L layers and H heads per layer maintains a residual stream $h^{(l)} \in \mathbb{R}^{T \times d}$, and the central identity (Elhage et al., 2021) is

$$h_t^{(L)} = h_t^{(0)} + \sum_{l=1}^{L} \sum_{h=1}^{H} head_{l,h}(h^{(l-1)})_t + \sum_{l=1}^{L} \mathrm{MLP}_l(h^{(l-1)})_t. \tag{1}$$

DLA projects this decomposition through a shared unembedding $W_U \in \mathbb{R}^{d \times |V|}$:

$$\ell_{v^+} - \ell_{v^-} = \sum_{l,h} head_{l,h} \cdot (W_U[:, v^+] - W_U[:, v^-]) + \dots \tag{2}$$

Equation (1) is the substrate; (2) the readout. Activation patching, attribution patching (Nanda, 2023), path patching (Goldowsky-Dill et al., 2023), sparse autoencoders, and linear probes all use one or both. Their guarantees inherit the guarantees of (1) and (2): a single residual sum, a single linear readout, and a homogeneous stack of layers.

4. Seven Properties That Don't Carry Over from LLMs to VLAs

A1. Single residual stream. (1) accounts for contributions from all heads and MLP layers. DLA, attribution patching, path patching, and SAEs all rely on this. Two VLA classes break the assumption. Type V (cross-attention fusion, e.g. RoboFlamingo) injects visual information through cross-attention sublayers that read directly from the vision encoder; their contribution appears nowhere in (1). Type IV (flow-matching, e.g. π_0) is worse: a separate action expert taps the backbone at multiple intermediate layers, so the backbone is not even self-contained.

A2. Shared unembedding for the output of interest. A single W_U projects the final hidden state into output space, which DLA, SAE feature interpretation, and logit-lens analysis all assume. Type I VLAs (RT-2, OpenVLA) preserve W_U by discretising each action dimension into 64 bins placed in the text vocabulary; Types II–IV (π_0, GR00T N1, Octo, diffusion heads) replace it with a regression MLP, a denoising network, or an ODE-integrated flow field — none reducible to a matrix multiply.

A3. Modules computing the same kind of thing throughout. Every layer of a standard transformer does the same operation on a single stream, so a tool that works at layer ℓ works elsewhere. A VLA is three coupled networks — vision encoder, language/planning backbone, action head — each representing information in a different geometry, and the action head is no longer a transformer layer at all (MLP, denoising U-Net, or flow-matching expert). Logit-lens curves and per-layer SAEs computed uniformly across the full VLA conflate three kinds of computation. Tools must be applied to one module at a time.

A4. Verifiable ground truth at the decision level. Correctness for an LLM is a single target token; ablations flip the top token or not. Robotic tasks admit multiple valid trajectories, and Type I VLAs predict action dimensions sequentially, so circuits for joint k attend to already-generated joints $1, \dots, k-1$ as conditioning. DLA on joint k mixes the vision-to-action circuit with the inter-dimension coordination circuit. Mitigation: run DLA on action token 1 only (no prior action context), and evaluate ablations in action space (KL, rollout success), not by token rank.

A5. Architectural regularity. A homogeneous stack means a uniform tool can walk it end-to-end. MoE and heterogeneous-expert architectures route different token types to different FFN experts; dual-model flow-matching designs run a second transformer in parallel. The same "head 17 of layer 12" index corresponds to different sub-computations depending on the routing decision, and per-layer attribution averages over sub-computations that cannot meaningfully be combined.

A6. Low superposition in the relevant subspace. Features of interest are approximately linearly separable; SAEs and linear probes recover them. Action subspaces encode continuous physical quantities — Cartesian positions, joint angles, gripper apertures — whose natural geometry is a low-dimensional continuous manifold, not a discrete categorical code. Discretising that manifold into 64 bins packs a smooth structure into a 64-token vocabulary, correlating embedding rows and densely superposing the residual-space representation.

A7. Single-pass determinism. One forward pass produces one next token, so causal attribution is single-pass. Diffusion VLAs decide over T denoising passes; flow-matching VLAs decide over an ODE integration; action-chunking VLAs emit K future motor commands per forward pass. A one-pass ablation measures a compounded multi-step effect, not a single-step causal contribution. The remedy is to run the analysis at each denoising step or chunk position and report a per-step circuit; the LLM toolkit does not do this out of the box.

Why a single-pass VLA is harder to interpret than an LLM. LLM next-token training and VLA behavioural cloning are the same supervised problem — both minimise per-step loss

$$\hat{\pi} \approx \arg\min_{\pi \in \Pi} \sum_{i,t} \text{loss}\big(s_t^{(i)}, a_t^{(i)}, \pi\big)$$

over expert state-action pairs — yet have very different test-time consequences. Both are autoregressive: the output at t becomes input at $t+1$, so a per-step error places the model in an input distribution it was not trained on, and the drift compounds (the *curse of horizon*).

Ross & Bagnell (2010) *Ross–Bagnell compounding-error bound* states that for a behavioural-cloning policy π with per-step error ε under the expert distribution,

$$J(\pi) - J(\pi^*) \;\leq\; \varepsilon H^2,$$

and the H^2 factor is tight for BC. A K-step action-chunking VLA (Zhao et al., 2023) amortises this to $\mathcal{O}(\varepsilon H^2/K)$, since decisions occur only at chunk boundaries.

The same bound warns us about mech-interp inaccuracies. A slightly-wrong DLA-derived steering vector or head ablation introduces a per-step interpretation error $\varepsilon_{\text{interp}}$ that accumulates polynomially on an LLM but quadratically on a robot. On OpenVLA at $H{=}100$–200, drift is bounded by $\mathcal{O}(\varepsilon_{\text{interp}} H^2)$, so $\varepsilon_{\text{interp}} \sim 10^{-2}$ already compounds to many bins by rollout's end; a $K{=}8$ chunking VLA absorbs an order of magnitude.

Rollout-level success on a non-chunking VLA does not certify per-step correctness (surviving heads can compensate), and rollout-level failure does not refute it. Single-pass VLAs are friendliest to the LLM toolkit but the architecture class where a "small" per-step error pays the highest downstream price.

Practitioner pitfalls (instrumentation analogues). Even on a VLA whose architecture passes A1–A7, two pitfalls surface at instrumentation time. **P1: Constrained-decoding subvocabulary.** Inference-time sampling is restricted to a small action token range (e.g. IDs 32000–32063 on OpenVLA). Logit-space tools that softmax over the full $|V| = 32{,}064$ vocabulary normalise the action-token mass by $\sim 32{,}000$ irrelevant text-token logits and report $P(\text{correct action}) \approx 0$ at every layer (a false null). Fix: restrict the softmax denominator to the action subspace. **P2: Embedding-level visual injection.** The Prismatic projector injects 256 visual patch features at the embedding level, before tokenisation. Tools that read `input_ids.shape[1]` measure 25 tokens (text only) and report 0% visual attention. Fix: read the true sequence length from `hidden_states[0].shape[1]` $= 281$.

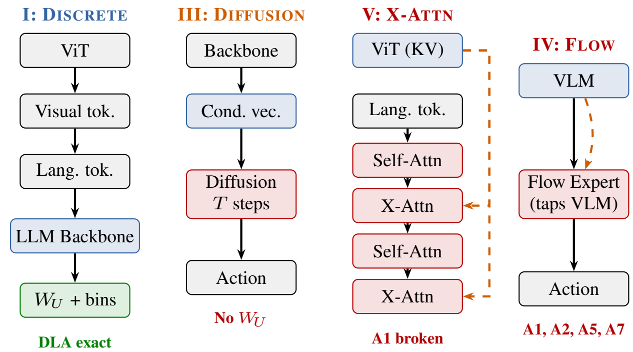

Figure 1. **Four representative VLA classes vs. A1 (single residual stream) and A2 (shared W_U).** Type I prepends visual tokens, keeping a single stream and a shared unembedding (DLA exact). Type III runs a separate diffusion head over T passes (no W_U, A7 violated). Type V injects vision via cross-attention sublayers (dashed orange) read at every other layer (A1 violated). Type IV runs a flow-matching action expert that taps multiple intermediate backbone layers (A1, A2, A5, A7 violated).

5. VLA Architecture Taxonomy

A VLA maps a visual observation o_t and a language instruction ℓ to an action a_t. The architecture divides into three components: vision encoder, language/action backbone, and action head. The interpretability surface is determined almost entirely by the action head type and the modality fusion strategy. We identify seven non mutually exclusive classes, illustrated for four representative cases in Figure 1.

Type I: Discrete-token VLAs (RT-2, OpenVLA). Pretrained LLM backbone, visual patch tokens prepended to language tokens, physical actions discretised into N bins per dimension as vocabulary tokens sharing W_U. A1 and A2 hold cleanly. A6 fails: the ordered action vocabulary correlates adjacent unembedding rows, densifying the action-block residual and frustrating SAE/SVD decompositions.

Type II: Continuous-regression VLAs (GR-1). Backbone identical to Type I; the action head is an MLP mapping the final hidden state to a continuous action vector. A2 fails (DLA needs a probe substitute). A6 fails downstream: continuous actions keep SAEs and SVD unreliable. Causal tools (path patching, activation steering) transfer.

Type III: Diffusion-based VLAs (Octo, Diffusion Policy). The backbone produces a conditioning vector; a separate diffusion network iteratively denoises an action from Gaussian noise. A1 fails (diffusion network sits outside the backbone residual stream); A2 fails (no W_U); A7 fails (T denoising passes, none of which is the decision).

Type IV: Flow-matching VLAs (π_0, $\pi_{0.5}$). A pre-trained VLM backbone and a separate action expert run simultaneously; the expert generates actions via flow-matching ODE integration, conditioning on intermediate backbone activations via cross-attention. A1 fails (action expert taps multiple backbone layers), A2 fails (no W_U on the action path), A5 fails (parallel networks break stack homogeneity), A7 fails (ODE distributes the decision over many steps).

Type V: Cross-attention fusion VLAs (RoboFlamingo). Visual features are not prepended; cross-attention sublayers query the vision encoder at every other layer. A1 fails structurally:

$$h_t^{(L)} = h_t^{(0)} + \sum_{l,h} head_{l,h} + \sum_{l} \underbrace{\mathrm{XA}_l(h^{(l)}, V_{\text{vis}})}_{\text{invisible to standard DLA}} + \dots \tag{3}$$

[1] If cross-attention carries fraction α of the action signal, every DLA score is deflated by $(1-\alpha)$. Standard tools without correction interpret a self-consistent shadow with the visual pathway removed.

Type VI: Heterogeneous-expert VLAs (π_0 action expert, SpatialVLA). Specialised sub-networks handle different modalities; in MoE variants, a router sends token types to different FFN experts. A5 fails (head index is no longer a single thing); A6 fails downstream. π_0 falls under both Type IV and Type VI — flow-matching and heterogeneous-expert decomposition are orthogonal.

Type VII: Temporal-chunking VLAs (ACT). The model predicts K future actions simultaneously; in ACT (Zhao et al., 2023) this is a conditional VAE with cross-attention at every decoder layer. A1 fails (chunked prediction violates single-stream accumulation), A4 fails (no single target token), A7 fails (decision distributed across the chunk).

6. How the Toolkit Breaks: Five Failure Modes

F1: Silent DLA underestimation (A1). If out-of-stream contributions carry fraction α of the action logit, every DLA score is deflated by $(1-\alpha)$. Within-stream rankings *survive* (heads deflated by the same factor preserve ratios), but absolute claims and ablation predictions do not: "head h accounts for 30% of the output" silently means "30% of $(1-\alpha)$", and ablations overstate the actual drop because the model routes around them through the unmeasured pathway.

F2: DLA against a causally disconnected readout (A2, Types II–IV). When no W_U exists for the action, substituting the text W_U runs without error but returns scores measuring contributions to a distribution the model does not use for control — ablating the top heads identified this way leaves rollout performance essentially unchanged. The valid substitute is a supervised probe, $\mathrm{ProbeDLA}_{l,h} = head_{l,h} \cdot w_{\text{probe}}$; it costs the elegance of training-free DLA but its numbers correspond to action-space behaviour.

[1] The elided terms are the standard MLP and embedding contributions of (1).

	A1	A2	A3	A4	A5	A6	A7
Type I	✓	✓	∼	✓	✓	×	✓
Type II	✓	×	✓	✓	✓	×	✓
Type III	×	×	✓	✓	∼	×	×
Type IV	×	×	✓	∼	×	×	×
Type V	×	×	∼	✓	×	×	✓
Type VI	✓	✓	✓	✓	×	×	✓
Type VII	×	×	✓	×	×	×	×

Table 1. LLM-side conditions for each VLA class. ✓ pass, × fail, ∼ partial. A1: single residual stream. A2: shared W_U. A3: homogeneity. A4: ground truth. A5: regularity. A6: low superposition. A7: single-pass.

F3: Variance-aligned direction-finding (A6). SVD picks directions of largest variance, but on robot data variance is the wrong objective: end-effector translation produces $\sim 100\times$ the variance of gripper aperture, so SVD's top components are translation-aligned and gripper-control axes land in tail components ($<1\%$ of variance). Heads ranked highest by SVD can be orthogonal to the variable they appear to identify, with near-zero causal effect. Supervised linear regression replaces SVD: it captures the variance that *predicts* the action, not the variance the residual happens to carry.

F4: Token-match evaluation hides action-space errors (A4). Standard mech interp judges a circuit by whether ablation flips the predicted token. On a discretised action vocabulary this is the wrong granularity: bin $31 \to 32$ flips the token but is behaviourally identical (both encode "slow right"), so an action-space-complete circuit can fail the ablation test; conversely, a circuit that preserves the token can still produce 5-bin shifts that compound into task failure.

F5: Heterogeneous-sequence ablation bias (A3, A6). The VLA token sequence mixes $\sim$256 visual, $\sim$25 language, and 1 action positions with very different activation statistics. Sequence-mean ablation substitutes a single value across all of them, biasing every head's importance by an OOD offset. Noise-based ablation is in-distribution at high-variance positions and wildly OOD at low-variance ones, so head rankings can flip with the noise scale chosen. Position-preserving mean ablation — $\mathbb{E}_{\text{dataset}}[x_{l,h,p}]$ at each position p separately — substitutes only values the model has seen at p, removing both biases.

7. Discussion

This paper does two things. First, it sets cautions for researchers transferring LLM mech-interp tooling and techniques to VLAs: the conveniences of the LLM toolkit are specific to one architectural choice, and most VLA architectures quietly violate them. Second, it argues that VLA mech interp is a fundamentally harder problem than its LLM counterpart, and motivates the new tooling that difficulty demands.

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